

Principles of Programming Languages

Lectures 5-6: Big-step Structural Operational Semantics

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Outline

Transition systems

Big-step SOS

Arithmetic expressions

Boolean expressions

Statements

Part 1

Transitions systems

- ▶ Operational semantics specifies how programs are executed
- ▶ How to define the operational semantics?
 - ▶ define a **transition system** which inductively specifies the step-by-step process of executing a program
- ▶ Transition systems are commonly used to model programs behavior

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 - ▶ define a **transition system** which inductively specifies the step-by-step process of executing a program
- ▶ Transition systems are commonly used to model programs behavior

Transition systems

- ▶ A **transition system** T is defined by:
 1. a set of **states** denoted $T(S)$ or S when T is understood from the context;
 2. a set of **initial states** denoted $T(I) \subseteq T(S)$;
 3. a set of **final states** denoted $T(F) \subseteq T(S)$;
 4. a transition relation $R \subseteq T(S) \times T(S)$

Transition systems

► Common notations and definitions

1. $s \Rightarrow s'$ means that $(s, s') \in R$;
2. if $s \in T(S)$ and there is no $s' \in T(S)$ such that $s \Rightarrow s'$ we say that s is *stuck*;
3. T is *deterministic* iff for every $s \in T(S)$ there is at most one $s' \in T(S)$ such that $s \Rightarrow s'$; otherwise, T is *non-deterministic*;
4. a *transition sequence* is a sequence of states $s_0, \dots, s_n$ such that $s_0 \in T(I)$ and $s_i \Rightarrow s_{i+1}$, for all $i \in \{0, n-1\}$;
5. a transition sequence $s_0, \dots, s_n$ is *maximal* if there is no $s' \in T(S)$ such that $s_n \Rightarrow s'$; if $s_n \in T(F)$ then $s_0, \dots, s_n$ is called *complete*.

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Reflexive and transitive closure

Notation for the reflexive and transitive closure of $\Rightarrow$:

$$s \Rightarrow^* s'$$

1. Reflexive: $s \Rightarrow^* s$;
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What is the induction principle corresponding to $\Rightarrow^*$?

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Big-step Structural Operational Semantics

Prerequisites: *states*

- ▶ The **state** is a *partial finite-domain function*:

$$\sigma : Var \rightarrow nat$$

- ▶ Example: $\sigma = x \mapsto 10, y \mapsto 1$

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State operations

► *Lookup*

$$_(_) : \textit{State} \times \textit{Var} \rightarrow \textit{nat}$$

► *Update*

$$_[_/_] : \textit{State} \times \textit{nat} \times \textit{Var} \rightarrow \textit{State}$$

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Big-step SOS

Ingredients:

► *Configuration*

- Tuple containing semantic ingredients
- Examples: $\langle a_1, \sigma \rangle$, $\langle a_1 + a_2, \sigma \rangle$, $\langle i_1 \rangle$, $\langle i_1 +_{nat} i_2 \rangle$, $\dots$

► *Sequent*

- Pair of configurations, to be *derived* or *proved*
- Examples: $\langle a_1, \sigma \rangle \Downarrow \langle i_1 \rangle$, $\langle a_1 + a_2, \sigma \rangle \Downarrow \langle i_1 +_{nat} i_2 \rangle$, $\dots$

► *Rule*

- Tells us how to derive sequents
- Example:

$$\frac{\langle a_1, \sigma \rangle \Downarrow \langle i_1 \rangle \quad \langle a_2, \sigma \rangle \Downarrow \langle i_2 \rangle}{\langle a_1 + a_2, \sigma \rangle \Downarrow \langle i_1 +_{nat} i_2 \rangle}$$

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 - ▶ Example: **premises**, **conclusion**, and **evaluates to**

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Rule schemata

- ▶ Rule schemata: **premises**, **conclusion**, and **evaluates to**

$$\frac{C_1 \Downarrow R_1 \quad C_2 \Downarrow R_2 \quad \cdots \quad C_n \Downarrow R_n}{C_0 \Downarrow R_0}$$

- ▶ You may also have met this terminology: inference rules, proof rules, deduction rules

Big-step rules for IMP – 1: arithmetic expressions

BIGSTEP-CONST: $\langle i, \sigma \rangle \Downarrow \langle i \rangle$

BIGSTEP-LOOKUP: $\langle x, \sigma \rangle \Downarrow \langle \sigma(x) \rangle$ if $\sigma(x) \neq \perp$

BIGSTEP-ADD:
$$\frac{\langle a_1, \sigma \rangle \Downarrow \langle i_1 \rangle \quad \langle a_2, \sigma \rangle \Downarrow \langle i_2 \rangle}{\langle a_1 + a_2, \sigma \rangle \Downarrow \langle i_1 +_{\text{nat}} i_2 \rangle}$$

BIGSTEP-MUL:
$$\frac{\langle a_1, \sigma \rangle \Downarrow \langle i_1 \rangle \quad \langle a_2, \sigma \rangle \Downarrow \langle i_2 \rangle}{\langle a_1 * a_2, \sigma \rangle \Downarrow \langle i_1 *_{\text{nat}} i_2 \rangle}$$

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Derivation

- ▶ A derivation example where $\sigma(n) = 10$:

$$\frac{\frac{\cdot}{\langle 2, \sigma \rangle \Downarrow \langle 2, \sigma \rangle} \text{ BIGSTEP-CONST} \quad \frac{\cdot}{\langle n, \sigma \rangle \Downarrow \langle 10, \sigma \rangle} \text{ BIGSTEP-LOOKUP}}{\langle 2 + n, \sigma \rangle \Downarrow \langle 2 +_{nat} 10, \sigma \rangle} \text{ BIGSTEP-ADD}$$

Part 2

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Big-step rules for IMP – 3: statements

BIGSTEP-ASSIGN:
$$\frac{\langle a, \sigma \rangle \Downarrow \langle i \rangle}{\langle x = a; , \sigma \rangle \Downarrow \langle \sigma[i/x] \rangle} \quad \text{if } \sigma(x) \neq \perp$$

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Big-step rules for IMP – 4: statements continued

BIGSTEP-WHILEFALSE:

$$\frac{\langle b, \sigma \rangle \Downarrow \langle \text{false} \rangle}{\langle \text{while } (b) \ s, \sigma \rangle \Downarrow \langle \sigma \rangle}$$

BIGSTEP-WHILETRUE:

$$\frac{\langle b, \sigma \rangle \Downarrow \langle \text{true} \rangle \quad \langle s \ \text{while } (b) \ s, \sigma \rangle \Downarrow \langle \sigma' \rangle}{\langle \text{while } (b) \ s, \sigma \rangle \Downarrow \langle \sigma' \rangle}$$

Big-step rules for IMP – 4: statements continued

$$\text{BIGSTEP-WHILEFALSE:} \quad \frac{\langle b, \sigma \rangle \Downarrow \langle \text{false} \rangle}{\langle \text{while } (b) \ s, \sigma \rangle \Downarrow \langle \sigma \rangle}$$

$$\text{BIGSTEP-WHILETRUE:} \quad \frac{\langle b, \sigma \rangle \Downarrow \langle \text{true} \rangle \quad \langle s \ \text{while } (b) \ s, \sigma \rangle \Downarrow \langle \sigma' \rangle}{\langle \text{while } (b) \ s, \sigma \rangle \Downarrow \langle \sigma' \rangle}$$

Exercise: add rule for division

► BIGSTEP-DIV:

$$\frac{\langle a_1, \sigma \rangle \Downarrow \langle i_1 \rangle \quad \langle a_2, \sigma \rangle \Downarrow \langle i_2 \rangle}{\langle a_1/a_2, \sigma \rangle \Downarrow \langle i_1/\text{nat } i_2 \rangle} \text{ if } i_2 \neq 0$$

► The following is a correct instance of BIGSTEP-DIV:

$$\frac{\langle x, (x \mapsto 10, y \mapsto 0) \rangle \Downarrow \langle 10 \rangle \quad \langle 2, (x \mapsto 10, y \mapsto 0) \rangle \Downarrow \langle 2 \rangle}{\langle x/2, (x \mapsto 10, y \mapsto 0) \rangle \Downarrow \langle 5 \rangle}$$

► What about the following?

$$\frac{\langle x, (x \mapsto 10, y \mapsto 0) \rangle \Downarrow \langle 8 \rangle \quad \langle 2, (x \mapsto 10, y \mapsto 0) \rangle \Downarrow \langle 2 \rangle}{\langle x/2, (x \mapsto 10, y \mapsto 0) \rangle \Downarrow \langle 4 \rangle}$$

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Big-step rules examples

- Recall BIGSTEP-DIV:

$$\frac{\langle a_1, \sigma \rangle \Downarrow \langle i_1 \rangle \quad \langle a_2, \sigma \rangle \Downarrow \langle i_2 \rangle}{\langle a_1 / a_2, \sigma \rangle \Downarrow \langle i_1 /_{\text{nat}} i_2 \rangle} \text{ if } i_2 \neq 0$$

- Is the following instance correct w.r.t. BIGSTEP-DIV?

$$\frac{\langle x, (x \mapsto 10, y \mapsto 0) \rangle \Downarrow \langle 10 \rangle \quad \langle y, (x \mapsto 10, y \mapsto 0) \rangle \Downarrow \langle 0 \rangle}{\langle x / y, (x \mapsto 10, y \mapsto 0) \rangle \Downarrow \langle ? \rangle}$$

- Answer: no

Big-step rules examples

- Recall BIGSTEP-DIV:

$$\frac{\langle a_1, \sigma \rangle \Downarrow \langle i_1 \rangle \quad \langle a_2, \sigma \rangle \Downarrow \langle i_2 \rangle}{\langle a_1 / a_2, \sigma \rangle \Downarrow \langle i_1 /_{nat} i_2 \rangle} \text{ if } i_2 \neq 0$$

- Is the following instance correct w.r.t. BIGSTEP-DIV?

$$\frac{\langle x, (x \mapsto 10, y \mapsto 0) \rangle \Downarrow \langle 10 \rangle \quad \langle y, (x \mapsto 10, y \mapsto 0) \rangle \Downarrow \langle 0 \rangle}{\langle x / y, (x \mapsto 10, y \mapsto 0) \rangle \Downarrow \langle ? \rangle}$$

- Answer: **no**

Derivation example

We assume that $\sigma \stackrel{\text{def}}{=} (x \mapsto 8, y \mapsto 0)$:

$$\frac{\text{BIGSTEP-LOOKUP} \frac{\cdot}{\langle x, \sigma \rangle \Downarrow \langle 8 \rangle} \quad \text{BIGSTEP-DIV} \frac{\text{BIGSTEP-LOOKUP} \frac{\cdot}{\langle y, \sigma \rangle \Downarrow \langle 0 \rangle} \quad \text{BIGSTEP-LOOKUP} \frac{\cdot}{\langle x, \sigma \rangle \Downarrow \langle 8 \rangle} \quad \text{BIGSTEP-LOOKUP} \frac{\cdot}{\langle 2, \sigma \rangle \Downarrow \langle 2 \rangle}}{\langle y/x, \sigma \rangle \Downarrow \langle 0 \rangle} \quad \text{BIGSTEP-INT} \quad \text{BIGSTEP-ADD} \quad \text{BIGSTEP-DIV} \frac{\langle x/(y/x + 2), \sigma \rangle \Downarrow \langle 4 \rangle}}{\langle y/x + 2, \sigma \rangle \Downarrow \langle 2 \rangle}}$$

Bibliography

- ▶ Chapter Simple Imperative Programs, Section: Evaluation as relation in **Software Foundations - Volume 1**, Benjamin C. Pierce, Arthur Azevedo de Amorim, Chris Casinghino, Marco Gaboardi, Michael Greenberg, Cătălin Hrițcu, Vilhelm Sjöberg, Andrew Tolmach, Brent Yorgey
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